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Anthropomorphic Style Construction Beyond Linguistic Signs in Generative
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Detecting Anthropomorphic Style in Language Models

Anthropomorphic Style Construction Beyond Linguistic Signs in Generative Language Models: A Rhetorical Perspective

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Abstract: This study addresses LLMs' deficiencies in decoding complex semantic relationships, particularly in identifying imagery-driven anthropomorphic rhetoric. We construct a six-dimensional taxonomy of personification constructs (role/designation, feature/part, emotion, action, relation, situational) through dual analytical lenses: single-sentence pattern recognition and compound-sentence contextual interpretation. The framework integrates exemplar-based semantic decomposition with syntactic dependency mapping, establishing formalized recognition patterns validated through automated detection experiments. Results confirm enhanced precision in anthropomorphic device identification and improved computational semantic parsing of rhetorical structures. The methodology provides standardized categorical criteria and annotated datasets for training domain-optimized models, while advancing systematic analysis of figurative language processing. This dual contribution enhances LLMs' capacity for stylistic discrimination in intelligent writing systems and establishes foundational protocols for digital rhetoric studies, facilitating interdisciplinary advancement in computational rhetoric and LLM-driven digital discourse analysis.

Keywords: Imagery Rhetoric; Personification Classification; Figure Recognition; Interpretation Patterns; Language Models

1. Introduction

1.1 Background and Problems

Large language models have had a profound impact on contemporary rhetoric, transforming the paradigms of rhetorical analysis, teaching, and research. They offer new avenues for study and possibilities for change, enabling rhetoricians to handle large datasets and conduct broader textual analyses (Cui Xiliang and Zhao Xia 2024). While these models have already developed general language rhetorical capabilities, they still exhibit certain limitations in understanding complex semantic relationships.

Rhetoric differs from grammar. Grammar structures describe the precise rules governing language structure and address whether a sentence is "correct" or not. On the other hand, rhetoric serves as methods or means to enhance linguistic expression, focusing on whether a sentence is "good" (Liu Huanhui 1991). The evaluation of how good an expression is depends on multiple criteria (Han Baocheng 2006). The quality of an expression relates to rhetoric, while the correctness of an expressive thought pertains to logic (Wu Liquan 2017). Although they address different scopes, the two are intricately connected internally. The three elements of language—pronunciation,

vocabulary, and grammar—are not only the means and foundation of rhetoric but also serve as rich linguistic materials that language users manipulate and process when employing rhetorical techniques (Chen Wangdao 1997,2008; Abdel-Raheem Ahmed 2024).

In studies on automatically identifying common rhetorical devices, methods based on model fusion for multi-rhetorical device recognition have shown lower accuracy rates in identifying metaphorical devices such as personification and metaphor. This indicates that relying solely on existing model-fusion-based methods is insufficient to effectively identify personification, metaphor, and other rhetoric types involving contextual meaning (referred to as contextual rhetoric) (Zhang Kai and Lv Xueqiang 2024). To improve recognition rates for such devices like personification and metaphor, alternative approaches are needed to thoroughly explore their internal semantic features and interpretative patterns (Jia Yuxiang and Yu Shiwen 2011). Compared to conventional rhetorical devices, contextual rhetoric extends beyond phonological, lexical, and grammatical levels, delving into the semantic level of sentences. In other words, the interpretative patterns of contextual rhetoric necessitate exploration at deeper semantic levels such as concepts and cognition (Skoufaki 2019).

1.2 Research Gap and Objectives

Contextual rhetoric is an important method for language users to communicate and cognize the objective world, and it is widely present in people's daily behaviors and lives. The extraction of semantic features and the exploration of interpretative patterns in contextual rhetoric are essential components of intelligent evaluations of Chinese rhetorical competence. The machine understanding process of contextual rhetoric (e.g., personification and metaphor) involves two stages: identification and evaluation. The first challenge to address is the recognition of contextual rhetoric. The ability to effectively identify contextual devices like personification and metaphor directly constrains numerous downstream tasks in natural language processing, including machine translation, question-answering systems, sentiment analysis, reading comprehension, human-computer dialogue, and automatic text summarization (Duan Caolin 2022).

The linguistic features of contextual rhetoric are not limited to phonological, lexical, or grammatical elements but also manifest in the understanding and application of abstract concepts at the cognitive level. Cognition refers to the process

by which individuals acquire or apply knowledge—the fundamental process of information processing in humans (Ausubel D P 2012). Contextual rhetoric can be seen as a process where language users further refine linguistic information based on their deep understanding of abstract concepts.

One crucial role of contextual rhetoric is to enable people to explore and perceive the external objective world or express inner thoughts and feelings. It achieves this by starting from existing concepts, actively observing independent phenomena, uncovering inherent connections between distinct entities, thereby adding new dimensions to established concepts. This represents not merely a quantitative change but a qualitative leap in understanding. To date, no mature semantic rules have been developed to systematically describe such features of contextual rhetoric (Hu Zhuanglin 1997).

2. Interpretation and Comparison of Personification in Contextual Rhetoric

Unlike conventional rhetorical devices with explicit/categorical categories such as parallelism, antithesis, repetition, rhetorical questions, and rhetorical questions, the identification of personification primarily relies on certain connections in the conceptual connotations or semantic combinations of sentence components rather than distinct grammatical or syntactic features (Zhang Gong 1963). It is important to note that the strength of these connections in conceptual connotations or semantic combinations within sentences directly impacts the effectiveness of the personification employed in such sentences (He Lianzhen 2019). Even among different sentences recognized as using personification, the impact of their semantic expressions can vary in intensity (Lin Yuwen 1959). As mentioned earlier, the linguistic features of personification within contextual rhetoric are not limited to aspects such as phonetics, vocabulary, and grammar, but also extend to cognitive patterns involving the understanding and application of abstract concepts. Therefore, exploring the interpretative principles of personification cannot be divorced from a cognitive model for its generation and identification.

The cognitive model of personification, grounded in the autonomous-dependency framework and integrating perceptual sensitivity and identity principles, explains its unique generation. As a pragmatic tool, personification's effectiveness relies on aligning rhetorical strategies with communicative goals. Tailored language adjustments are essential: simpler forms (iconic or role-based personification) suit young children with limited knowledge,

while complex forms (emotional or relational personification) better engage university students with advanced cognitive reserves, ensuring audience-appropriate conceptual resonance.

To facilitate discussion of the fundamental principles underlying the cognitive models for personification generation and recognition, we propose two terms from a semantic analysis perspective: "ant" and "be-anted item". These terms are defined as follows:

Definition 1: An ant refers to a semantic unit with anthropomorphic characteristics in a personified expression.

Definition 2: A be-anted item refers to a non-human entity that is attributed with human characteristics in a personified expression.

Here, "ant" corresponds to the concept of "antecedent" proposed by Professor Zhou Changle, and "be-anted item" corresponds to the concept of "antitype" (Zhou Changle 2021). Take the following two example sentences as illustration:

(1) The east wind has arrived; spring's footsteps are near. (From Zhu Ziqing "Spring")

(2) The candle quietly weeps. (From the Eighth Grade Chinese textbook "The Candle")

"spring" is a non-human entity, and "footsteps" represents human characteristics. Therefore, "spring" serves as the be-anted item ("antitype"), and "footsteps" serves as the ant ("antecedent"), see (1).

"candle" is a non-human entity, while "weeping" represents a human characteristic. Thus, "candle" acts as the be-anted item ("antitype"), and "weep" functions as the ant ("antecedent"), see (2).

From the examples above, it can be observed that the generation of personified expressions is triggered by the compositional intention of the discourse subject. The discourse subject performs semantic analysis and processing of "be-anted items" and "ants" guided by cognitive orientation, to identify similarities between these two elements and further develop, refine or adjust them through inferential processes. Simultaneously, guided by the linguistic rules governing personification, the discourse subject integrates syntactic components with elements of the referenced objects,

ultimately achieving the expression and generation of a personified sentence (Figure 1).

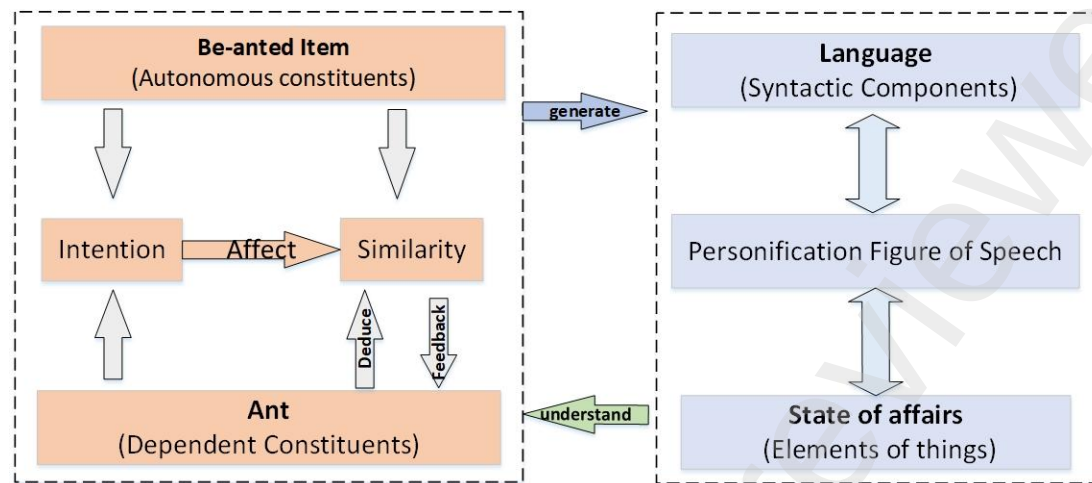


Figure 1 Cognitive Model for Personification Text Generation and Recognition

Compared to the generation process of personified expressions, the recognition of such sentences is precisely the reverse of the generation process: Under reasonable syntactic component segmentation and thorough analysis of language expression rules for personification and corresponding conceptualizations of objects, similarities between "be-anted items" and "ants" are sought. Through the inverse deconstruction of cognitive orientation, Semantic restoration is performed on the "be-anted items" and "ants" processed by the discourse subject. This ultimately identifies the compositional intention that triggered the discourse subject to employ personification, thereby recognizing its use.

3 The classification system of anthropomorphic artistic conception rhetoric oriented towards understanding

In Chinese statements, the use scope of anthropomorphic rhetoric is very wide and appears in large numbers in fairy tales, novels, poems and other literary works, presenting rich expression forms. Based on the generation model of anthropomorphic rhetorical sentences and the recognition model of anthropomorphic rhetorical sentences mentioned in the previous section, this section gives a classification system of anthropomorphic artistic conception rhetoric oriented towards understanding from the perspectives of single-sentence anthropomorphic recognition and complex-sentence anthropomorphic recognition. Among them, single-sentence anthropomorphic recognition mainly includes six major categories such as identity/appellation anthropomorphism, feature/part anthropomorphism, emotional

anthropomorphism, action anthropomorphism, relationship anthropomorphism, and event anthropomorphism. Complex-sentence anthropomorphic recognition mainly includes two major categories such as parallel sentence anthropomorphism and clause anthropomorphism. Anthropomorphic bases, anthropomorphic objects and similarities determine cognitive structures of anthropomorphic rhetoric. By giving examples of different types of anthropomorphic rhetoric, semantic analysis and dependency syntactic analysis diagrams, the framework of different types of anthropomorphic patterns is determined and the formal expression of recognition rules can be given.

3.1 Definition

Set:

$N_{\text{non-human}}$ indicates the set of non-human nouns.

N_{identity} indicates the set of identity nouns.

$N_{\text{human exclusive relationship}}$ indicates the set of noun phrases specifically used to represent human exclusive relationships.

$N_{\text{human exclusive event word}}$ indicates the set of noun phrases specifically used to represent human exclusive events.

Q_{emotion} indicates the set of word phrases specifically used to represent human emotions.

$S_{\text{identity personification}}$ indicates the set of identity personification rhetorical sentences.

$S_{\text{partial personification}}$ indicates the set of single rhetorical sentences with feature/partial personification.

$S_{\text{action personification}}$ indicates the set of single rhetorical sentences with action personification.

$S_{\text{emotion personification}}$ indicates the set of single rhetorical sentences with emotion personification.

$S_{\text{relationship personification}}$ indicates the set of single rhetorical sentences with relationship personification.

$S_{\text{event personification}}$ indicates the set of single rhetorical sentences with event personification.

S indicates the set of sentences.

DP indicates the set of dependency syntax trees of S .

VX indicates the set of system words.

Hs indicates the set of all words/phrases in sentence s .

Variables:

Noun variables $n_1, n_2, n_3, \dots$

Verb variables $v_1, v_2, v_3, \dots$

Emotion word variables $q_1, q_2, q_3, \dots$

Sentence variables $s_1, s_2, s_3, \dots$

Relationship:

$x+y$ indicates that x and y appear in a sentence one after another, and there are no other sentence components between them. Here, x and y can be any of the above variables.

3.2 Construction of single-sentence interpretation mode

Drawing on Professor Zhou Changle's relevant analyses of metaphor computing, our research starts from the prototypes, analogs and similarity relationships in personification rhetoric. By analyzing the cognitive structures of the tenor, the vehicle and the similarity, six single-sentence personification rhetorical interpretation models are obtained (as shown in Table 1). And on the basis of Professor Zhou Changle's metaphor classification marking symbol system, the personification classification system marking instructions are formed (Zhou Changle 2021)(as shown in Table 2).

Table 1 A cognitive structure classification result table of Chinese single-sentence personification rhetoric

NO.	Name	Feature Description	Example
1	Single-Sentence with Identity/Appellation Personification	The original and the imitated entity exist side by side and jointly hold a certain component of the sentence. The original is directly regarded as a certain kind of person represented by the imitated entity.	(1) The baby elephant left Ruerai Prairie. (2) Corn is the breakfast of the baby elephant.
2	Single-Sentence with Features/Partial Personification	There is a similarity in certain characteristics between the (part of) the original and the (part of) the imitated entity.	(1) The words of the seagull are filled with joy. (2) The past of the seagull is full of loneliness.
3	Single-Sentence with emotional Personification	There is a significant similarity in emotions between the original and the imitated entity.	(1) The ashamed fox left the forest. (2) The little fish swam back and forth under the lotus leaf with a smile. (3) Autumn walked towards us steadily with slow steps. (4) The seagull cast a meaningful glance at the sky.
4	Single-Sentence with action Personification	There is a significant similarity in the core actions between the original and the imitated entity.	(1) The corn that the elephant forgot in the warehouse last year became the breakfast of the mice. (2) The butterfly is dancing gracefully among the flowers. (3) The little seagull studies very hard.
5	Single-Sentence with relational Personification	Directly regard a certain relationship among multiple original texts as a certain relationship peculiar to human beings.	(1) The little pig, the little sheep and the little ducks are all good friends. (2) The fox and the earth dog are friends. (3) This group of elephants is the most fierce legion on the grassland. (4) The cat and the tiger are master-apprentice relationship.
6	Single-Sentence with event Personification	Directly regard a certain relationship among multiple simulated originals as a certain relationship peculiar to human beings.	(1) The seagulls are holding a grand dance party. (2) This is a collective suicide of blue whales.

Table 2 Marking Explanations of Personification Classification System (Zhou Changle 2021)

a Adjective	r Pronoun	LAD Pre-appendage relationship
b Distinctive word.	u Auxiliary word	ARD Post-appendage relationship
c Conjunction	v Verb	VV Serial verb construction
d Conjunction	vb Auxiliary verb	ADV Adverbial-Modifier structure
e Interjection	vd Verbal Adverb	CNJ Correlative structure
g Morpheme character	vf Form verb	IS Independent structure
h Preceding component	vg General verb	MT Voice structure
i Idiomatic expression	vn Verbal noun	CMP Verb-Complement structure
j Abbreviation	vq Directional verb	DE construction
k Following component	vx Linking verb	DI construction
m Number word	vz Modal verb	DEI construction
n Noun	wp Punctuation	BA construction
nd Directional noun	ws String	BEI construction
nh Personal name	x Non-morpheme character	HED core
ni Organizational proper name	ATT Attributive-modifier relationship	IC Independent clause
nl Locative noun	COO Coordinative relationship	DC Dependent clause
ns Geographical name	VOB Verb-object relationship	POB Prepositional-object relationship
nt Temporal noun	APP Appositive relationship	SBV Subject-predicate relationship
nz Other proper name	p Preposition	SIM Similar relationship
o Onomatopoeia	q Measure word	<EOS> Period mark

3.2.1 Single-Sentences with Identity/Appellation Personification

Through semantic analysis, single-sentence with identity/appellation personification exhibit the following prominent features:

Core human-identity terms: These sentences contain nouns denoting human identities (titles or professions) as core words, with non-human nouns serving as modifiers.

Flexible syntactic positions: The "modifier-head" phrases can appear flexibly in subject, object, or attributive positions. **Hybrid structure:** Combines non-human entities with human-specific roles or titles. Examples of personified identity rhetorical

sentences:

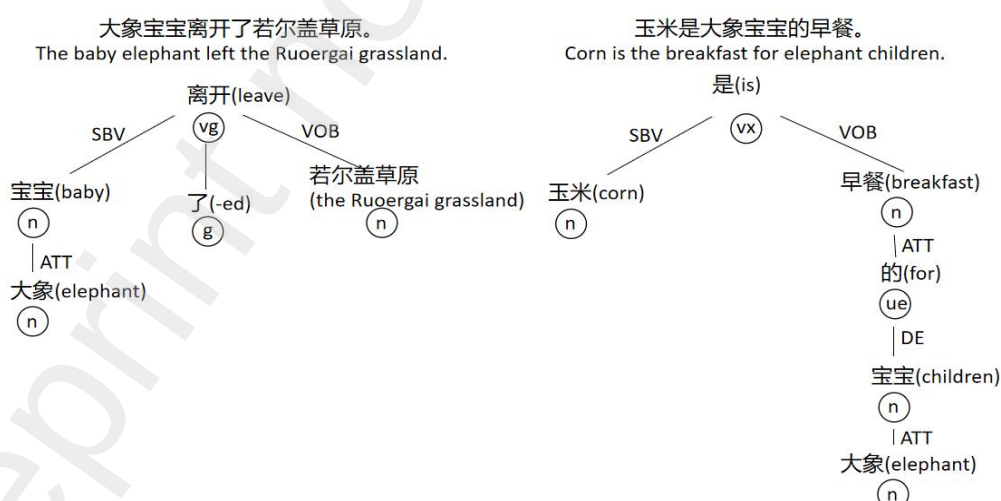
(3) *The baby elephant left the Ruorgai grassland.*

(4) *Corn is the breakfast of the elephant baby.*

(5) *No one can convince the fox lawyer.*

(6) *This elephant is still a baby.*

To analyze this type of anthropomorphized rhetorical single-sentence, we first need to identify expressions like "elephant baby" or "fox lawyer," then recognize the phenomenon where pronouns like "he" "she" "they" or "them" refer to non-human nouns. First, when a phrase appears in the form of " $N_{\text{non-human}} + N_{\text{human-exclusive identity}}$ " as a possessive structure, it is considered anthropomorphism. Second, when the sentence follows a "subject-linking verb-predicate" structure, where the subject is a non-human noun and the predicate is a human-exclusive identity noun, it is also considered anthropomorphism. Human-exclusive identity nouns include terms that refer to professions (such as lawyer, doctor, etc.) and terms that refer to specific human titles (such as baby, grandpa, etc.). The dependency syntax analysis diagrams of anthropomorphized rhetorical single-sentences are as follows:



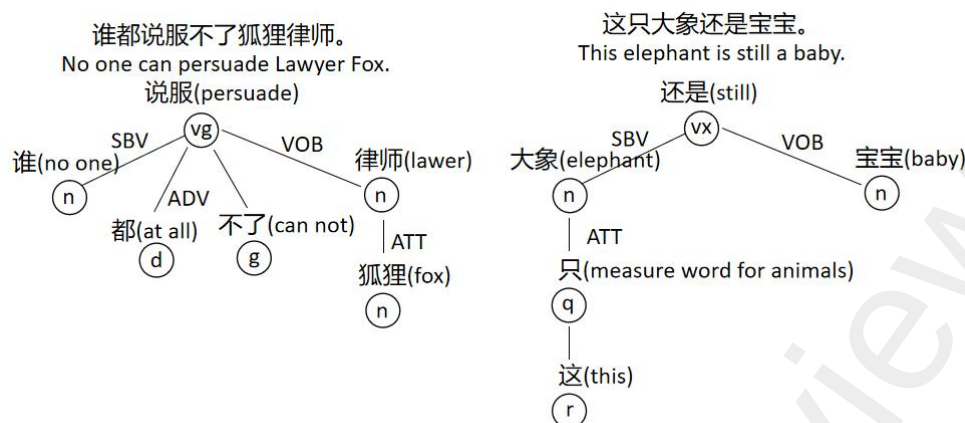


Figure2 Dependency Syntax Analysis figure for anthropomorphized rhetorical single-sentences

In (3)-(6) above, whenever the combination " $N_{\text{non-human}} + N_{\text{identity}}$ " appears, it can be identified as an anthropomorphized rhetorical phenomenon of identity. This feature is a semantic characteristic at the lexical level and is relatively easy to capture. As a special sentence structure, the subject-predicate-complement structure has a distinct predicate feature (linking verb). From the dependency syntactic analysis diagram of the example sentences, The pattern framework of single-sentence with identity/appellation personification is as follows:

Table 3 Semantic pattern Table of Single-Sentences with Identity/Appellation

Category	Description	Framework	Example Sentence
1	"x ATT y" structure: x is an identity noun, y is an animal noun.	宝宝(baby) (n) ATT 大象(elephant) (n)	The baby elephant left the Ruorgai grassland.
2	Subject-predicate-complement structure: the subject is a non-human noun, and the predicate nominative is a human-exclusive identity noun.	还是(still) (vx) / SBV \ VOB 大象(elephant) 宝宝(baby) (n) (n)	This elephant is still a baby.

Rule 1: If $\exists n_1 \in N_{\text{non-human}}, \exists n_2 \in N_{\text{identity}}, \exists s \in S, n_1+n_2 \in Hs$ or $n_2+n_1 \in Hs$, then $s \in S_{\text{anthropomorphized identity}}$.

Rule 2: If $\exists n_1 \in N_{\text{non-human}}, \exists n_2 \in N_{\text{identity}}, \exists s \in S, \exists v \in VX, n_1 \in Hs, n_2 \in Hs$, the dependency structure syntactic analysis of s $DP_s = N_{\text{non-human}} + VX + N_{\text{identity}}$, then $s \in S_{\text{anthropomorphized identity}}$.

3.2.2 Single-Sentences with Features/Partial Personification

The most prominent semantic feature of single-sentences with features/partial personification is that some characteristics or part of the entity being personified share similarities with the entity being personified. Compared to single-sentences with

identity/appellation personification, the similarity between the personified item and the entity being personified is weaker in single-sentences with features/partial personification. These types of anthropomorphized rhetorical single-sentences have a more flexible sentence structure, often including a noun phrase formed by the personified item and the entity being personified, which can serve as the subject of the single-sentences with features/partial personification, or as the object of the sentence. Additionally, there is the "subject-linking verb-predicate" sentence structure where the be-anted item being personified serves as the subject of the sentence and the ant serves as the predicate. Examples of single-sentences with features/partial personification are as follows:

(7) *The seagull's words are full of philosophy.*

(8) *This book records the glorious history of the saber-toothed tiger.*

(9) *The butterfly has an elegant posture.*

To identify Single-Sentences with Features/Partial Personification, it is necessary to first recognize semantic expressions such as “the seagull's words” and “the glorious history of the saber-toothed tiger”. For sentences that include a noun phrase with a modifier structure, a combination of "a word expressing a non-human entity + a noun indicating a human-specific part" in the sentence is considered a single-sentences with features/partial personification. For sentences with the "subject-linking verb-predicate" structure, if the subject is a non-human noun and the predicate is a noun that refers to a human-specific part, it is also identified as a single-sentences with features/partial personification. The dependency syntax analysis diagrams for single-sentences with features/partial personification are as follows:

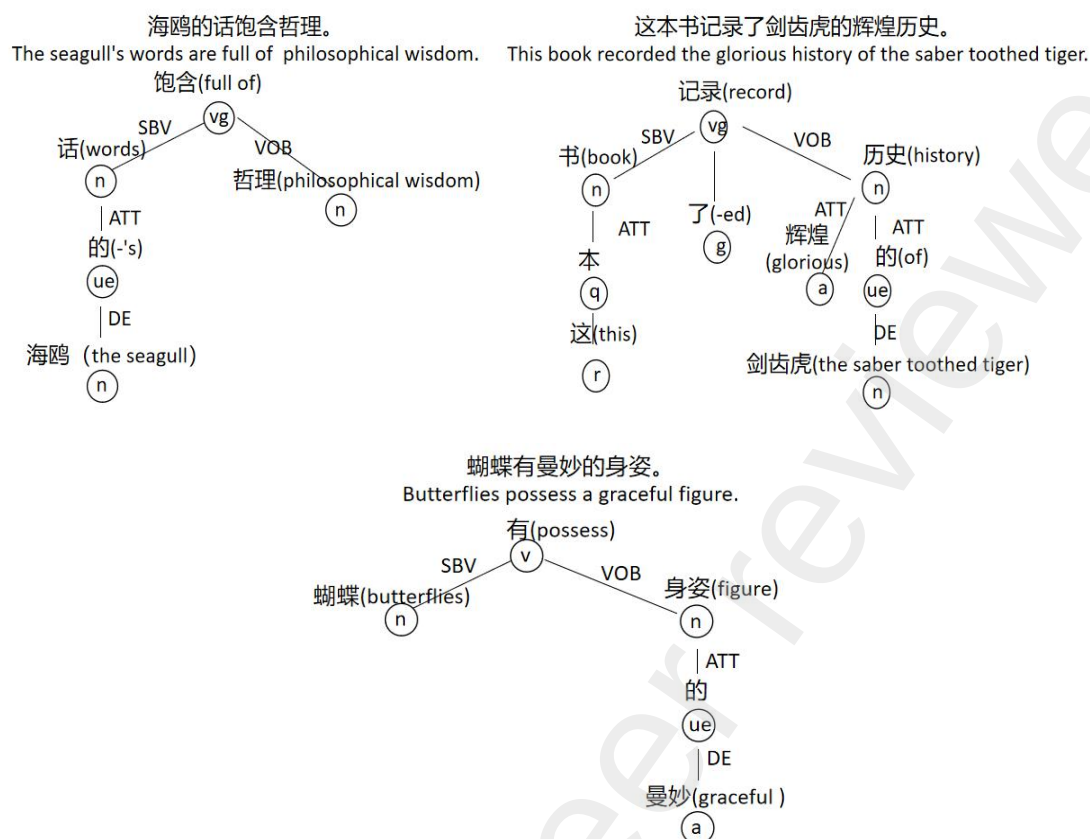


Figure 3 Dependency Syntax Analysis figure for Single-Sentences with Features/Partial Personification

In (7) and (8), regardless of the grammatical level at which the noun phrase with the structure “ $N_{\text{non-human}} + N_{\text{human-specific part}}$ ” appears, it is identified as feature/partial anthropomorphized. In Example 9, when the sentence consists of the structure “ $N_{\text{non-human}} + \text{linking verb} + N_{\text{human-specific part}}$ ” forming the "subject-linking verb-predicate" structure, it is identified as feature/partial anthropomorphized. That is to say, whenever this combination appears in a single-sentence, it is considered a single-sentences with features/partial personification. This semantic feature no longer appears at the lexical level, but tends to appear at the phrase or sentence level. Therefore, the identification of single-sentences with features/partial personification are more complex. Based on the dependency syntax analysis of single-sentences with features/partial personification, the The pattern framework of single-sentences with features/partial personification are summarized.

Table 4 Semantic pattern Table of Single-Sentences with Features/Partial Personification

Category	Description	Framework	Example Sentence
1	"x ATT y" Structure: x is a non-human noun, and y is a noun exclusive to humans.	话(words) (n) ATT 的('s) (ue) DE 海鸥(seagull) (n)	Seagulls' words are filled with philosophy.

2	Subject-Linking Verb-Predicate Structure: The subject is a non-human noun, and the predicate is a noun exclusive to humans.		The butterfly has a graceful posture.
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Rule 3: if $\exists n_1 \in N_{\text{non-human}}, \exists n_2 \in N_{\text{human-exclusive part}}, \exists s \in S, n_1 \in Hs, n_2 \in Hs, N_{\text{non-human}}$ and $N_{\text{human-exclusive part}}$ formed a ATT structure, then $s \in S_{\text{human-exclusive part}}$.

Rule 4: if $\exists n_1 \in N_{\text{non-human}}, \exists n_2 \in N_{\text{human-exclusive part}}, \exists s \in S, n_1 \in Hs, n_2 \in Hs, N_{\text{non-human}}$ and $N_{\text{human-exclusive part}}$ form a subject-predicate-copula structure, then $s \in S_{\text{human-exclusive part}}$.

3.2.3 Single-Sentence with Action Personification

Single-sentence with action personification is most notably characterized by the semantic feature that the action of the personified element and the human-specific action share significant similarity, or the action of the personified element is directly perceived as an action exclusive to humans. In such sentences, the distinction between the ant and the be-anted item within the sentence structure is often not explicitly clear. The sentence may give the impression of containing only the personified element without a clearly defined personifying element. Syntactically, these sentences typically form a subject-predicate phrase structure (composed of the personified element and the action), which may either stand alone as a complete sentence or function as a component within a larger sentence. Examples of single-sentences with action personification include:

(10) *The butterfly danced gracefully among the flowers.*

(11) *The little seagull studies diligently.*

(12) *The corn that the elephant forgot in the warehouse last year became breakfast for the mice.*

To identify a single-sentence with action personification, first recognize human-specific verbs (e.g. "dance gracefully" "study diligently" or "forget") that are typically exclusive to human actions. For such sentences, if the syntactic structure contains a subject-predicate phrase where the subject is a non-human noun and the predicate is a human-specific verb, it qualifies as action personification. Below are

examples of dependency syntactic analysis diagrams for single-sentences with action personification:

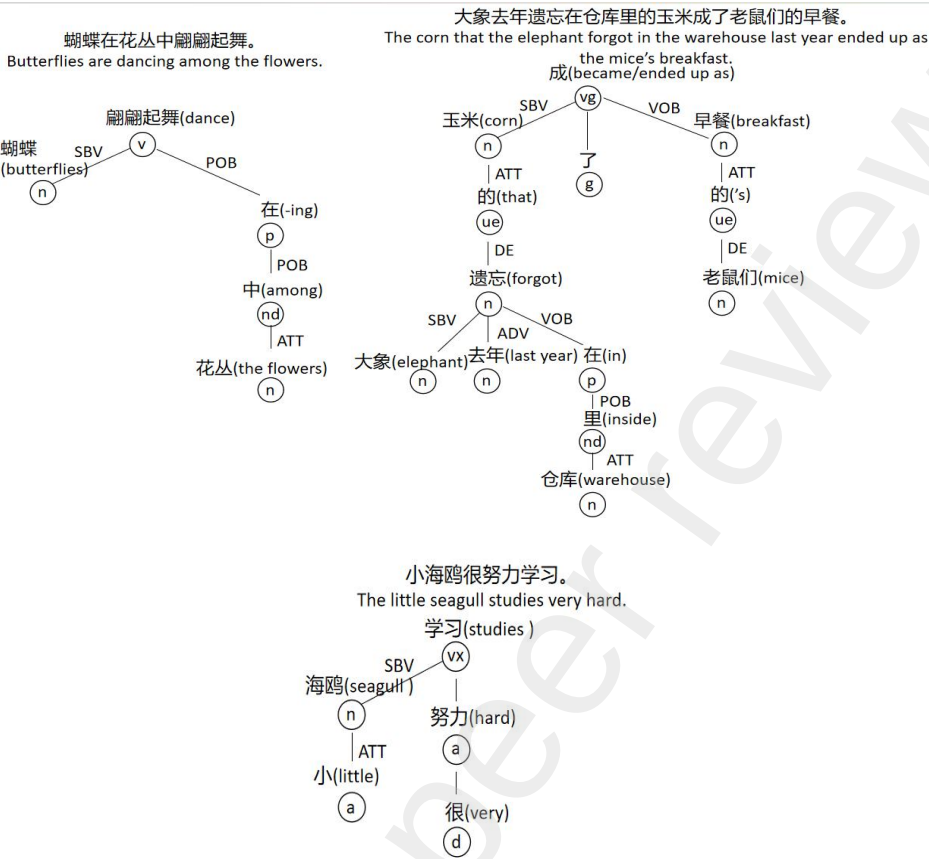


Figure 4 Dependency Syntax Analysis figure for Single-Sentences with Action Personification

In the examples (10)-(12), whenever the combination of “N_{non-human} + V_{human-exclusive}” appears at any grammatical level, it is identified as action personification. In other words, the presence of this structural pattern alone qualifies it as a personification device. Compared to single-sentences with feature/partial personification, the semantic characteristics of single-sentences with action personification remain primarily at the lexical level (centered on verbs), making them relatively straightforward to extract. As demonstrated by the dependency parsing diagrams of the example sentences provided, The pattern framework of action personification follows the structure below:

Table 5 Semantic pattern Table of Single-Sentences with Action Personification

Category	Description	Framework	Example Sentence
1	Structure "x SBV y": x is a N _{non-human} , y is a V _{human-exclusive} .	遗忘(forgot) (n) SBV 大象(elephant) (n)	The corn that the elephant forgot in the warehouse last year became breakfast for the mice.

Rule 5: if $\exists n_1 \in N_{\text{non-human}}, \exists v_1 \in V_{\text{human-exclusive}}, \exists s \in S, n_1 \in H_s, v_1 \in H_s,$
 $N_{\text{non-human}}$ and $V_{\text{human-exclusive}}$ has an SBV (Subject-Verb) relationship, then $s \in S_{\text{action}}$
 personification.

3.2.4 Single-Sentence with emotion Personification

The prominent feature of single-sentences with emotion personification is the appearance of emotional words. It is generally believed that emotion is a unique attribute of human beings. Therefore, when emotional words are used to modify the core words of animal nouns, the phenomenon of emotional anthropomorphic rhetoric appears. Due to the rich types of emotional words (such as adjectives, adverbs, verbs, nouns, idioms, prepositions), the positions of emotional words in sentences often change, which leads to the more complex semantic characteristics of single-sentence with emotion personification. Examples of single-sentences with emotion personification are as follows:

(13) *The water licks people's waists wantonly.*

(14) *The ashamed fox left the forest.*

(15) *The little fish swims back and forth happily under the lotus leaf.*

(16) *Autumn steps steadily and slowly towards us.*

(17) *The little rabbit is very ashamed.*

To identify single-sentences with emotion personification, it is necessary to first identify emotional expressions or directly emotional words. Given that the semantic characteristics of single-sentences with emotion personification are relatively complex, the identification rules are as follows:

In a complete sentence that does not nest subject-predicate phrases, if the subject of this structure is a non-human noun and the emotional word appears in this subject-predicate phrase (or the subject in the smallest subject-predicate structure where the emotional word is located is a non-human noun), it is determined that there is an emotional anthropomorphic rhetorical phenomenon, and the sentence is a single-sentence with emotion personification.

When a non-human noun is the core word and the emotional word acts as a modifier of the core word, it is determined that there is an emotional anthropomorphic phenomenon (the non-human noun can be the subject or the object), and the sentence is a single-sentence with emotion personification.

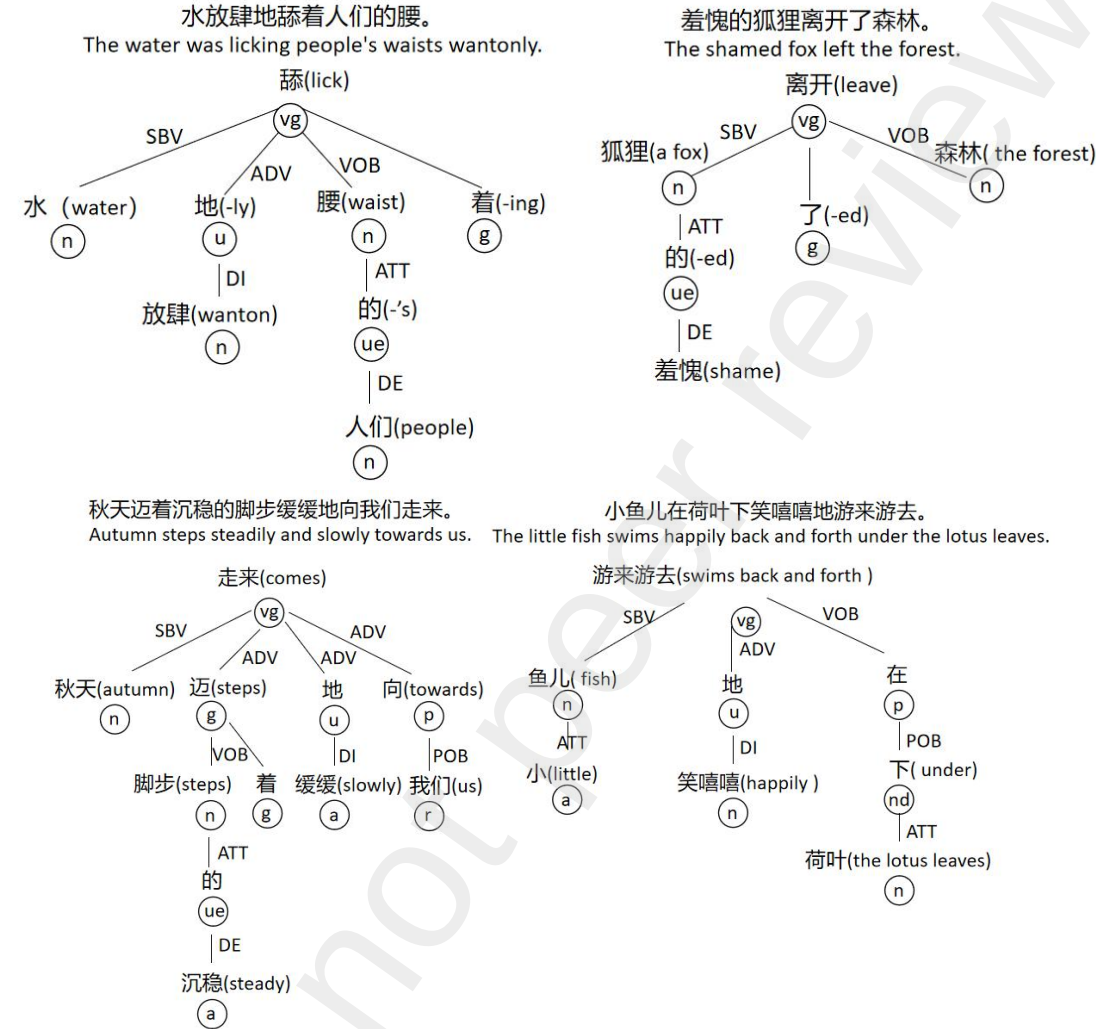


Figure 5 Dependency Syntax Analysis figure for Single-Sentences with emotion Personification

In single-sentences with emotion personification, no matter where the emotional word “Q_{emotion}” appears in the grammatical hierarchy, and no matter whether it appears in the adverb position (such as (2) and (3)) or the predicate position (such as (1)), as long as the subject in the smallest subject-predicate structure where the emotional word is located is a non-human noun, it is recognized as an emotional anthropomorphic rhetorical phenomenon. The pattern framework of single-sentences with emotion personification are as follows:

Table 6 Semantic pattern Table of Single-Sentences with emotion Personification

Category	Description	Framework	Example Sentence
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1	Structure “x ATT y”: x is a $Q_{emotion}$, y is a $V_{non-human}$.	狐狸(a fox) (n) ATT 的(-ed) ue DE 羞愧(shame)	The ashamed fox left the forest.
2	Structure “x SBV y ADV *DI z”: x is a $N_{non-human}$, y is a Verb, and z is a $Q_{emotion}$.	游来游去(swims back and forth) SBV 鱼儿(fish) (n) ATT 小(little) (a) ADV 地 u DI 笑嘻嘻(happily) (n)	The little fish swims back and forth happily under the lotus leaf.
3	Structure “x SBV y”: x is a $N_{non-human}$, y is a $Q_{emotion}$.	惭愧(ashamed) SBV 兔子(rabbit) (n)	The little rabbit is very ashamed.

Rule 6: if $\exists q_1 \in Q_{emotion}$, $\exists n_1 \in N_{non-human}$, $\exists s \in S$, $q_1 \in H_s$, $n_1 \in H_s$, q_1 the core word in the smallest attributive-head structure (ATT) where it is located is n_1 , then $s \in S_{emotional personification}$.

Rule 7: if $\exists q_1 \in Q_{emotion}$, $\exists n_1 \in N_{non-human}$, $\exists s \in S$, $q_1 \in H_s$, $n_1 \in H_s$, and q_1 the subject in the smallest subject-predicate structure (SBV) where it is located is n_1 , then $s \in S_{emotional personification}$.

Rule 8: if $\exists q_1 \in Q_{emotion}$, $\exists n_1 \in N_{non-human}$, $\exists s \in S$, $q_1 \in H_s$, $n_1 \in H_s$, and the subject of the sentence q_1 is n_1 , then $s \in S_{emotional personification}$.

3.2.5 Single-Sentence with relational Personification

The distinctive feature of single-sentence with relational personification is that the relationships between the items being personified exhibit considerable similarity, or in other words, the relationships between be-anted item can be seen as a specific relationship between ant¹. Unique relationships between individuals are represented using nouns that denote interpersonal relationships.

Although there exists a multitude of relationships among individuals², the positioning of nouns that denote interpersonal relationships in sentences is usually not complex and tends to be relatively fixed. From a syntactical perspective, it primarily comprises a "subject-linking verb-predicate" structure formed by the ant and be-anted item (I.A. Richards 1936) the subject is a non-human noun, while the predicate is a

¹ The term "items being personified" is not entirely clear here. The relationship between the items can be the relationship illustrated in example 1, or it can be the relationship shown in example 3. However, it is evident that in example 3, the search and rescue dogs and humans are not all personified entities; it would be more accurate to describe this as the relationship between the personified entity and a third party.

² Such as teacher-student relationships, classmate relationships, subordinate relationships, colleague relationships, partner relationships, friendships, romantic relationships, business relationships, spousal relationships, mother-child relationships, father-child relationships, grandparent-grandchild relationships, and so forth.

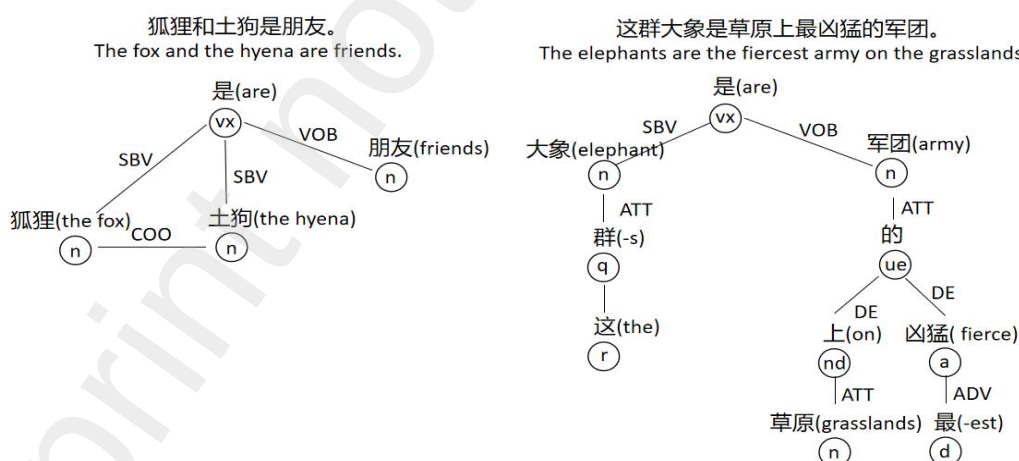
noun representing an interpersonal relationship.

(18) *The fox and the stray dog are friends.*

(19) *This group of elephants is the fiercest legion on the prairie.*

(20) *The cat and the tiger are in a master-apprentice relationship.*

To identify single-sentence with relational personification, one must first recognize expressions of interpersonal relationships. In a complete "subject-linking verb-predicate" structure without nested subject-verb phrases, if the subject of the structure consists of two or more non-human nouns and an interpersonal relationship noun appears in the predicate position, it is considered that a phenomenon of relational personification exists. If the subject of the structure is a single non-human noun, and the interpersonal relationship noun appears in the predicate position, and this noun also serves as the core term in a definitional structure³, it likewise indicates the presence of relational personification, thus categorizing the sentence as a single-sentence with relational personification. The syntactic dependency analysis diagram for examples of single-sentences with relational personification are as follows:



³ Where the modifying part can be either a non-human noun or a noun denoting an interpersonal relationship, as seen in (19) and (20).

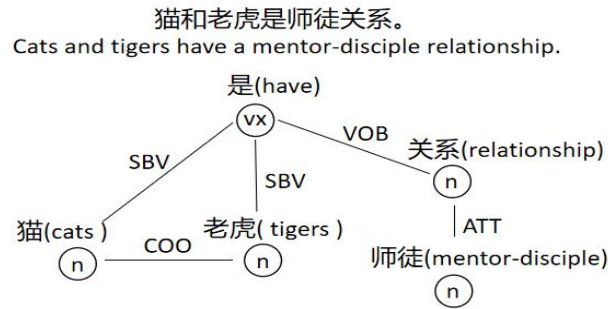


Figure 6 Dependency Syntax Analysis figure for single sentences with Relational Personification

In the examples of single-sentences with relational personification, regardless of the semantic structure, the interpersonal relationship noun $N_{\text{interpersonal relationship}}$ appears in the predicate position of the sentence. As long as the subject in the minimal subject-verb structure containing the interpersonal relationship noun is a non-human noun, the presence of the interpersonal relationship noun is recognized as indicating a phenomenon of relational personification, thus categorizing the sentence as a single-sentence with relational personification.

From the syntactic dependency analysis diagram of relational personification rhetoric sentences (Figure 6), we can derive The pattern framework of the relational personification mode as follows:

Table 7 Semantic pattern Table of Single-Sentences with Relational Personification

Category	Description	Framework	Example Sentence
1	"Subject-Verb-Predicate" structure: The subject is a non-human noun, and the predicate is an interpersonal relationship noun.		This group of elephants is the fiercest legion on the grassland.

Rule 9: if $\exists n_1 \in N_{\text{person specific relationship}}$, $\exists n_2 \in N_{\text{non-human}}$, $n_1 \in H_s$, $n_2 \in H_s$, n_1 is in the subject-predicate-copula sentence structure, it serves as the predicative. n_2 serves as the subject of the sentence, then $s \in S_{\text{relationship personification}}$.

3.2.6 Single-Sentence with event Personification

The characteristic of single-sentence with event personification focused on events is that the events participated in or engaged in by the subject of personification show a significant similarity to events unique to the subject, or in other words, the events in which the subject participates or engages are directly viewed as an event that is particular to the subject (human). The expression of events unique to humans is not singular; some are represented using specific event nouns (proper nouns), while others

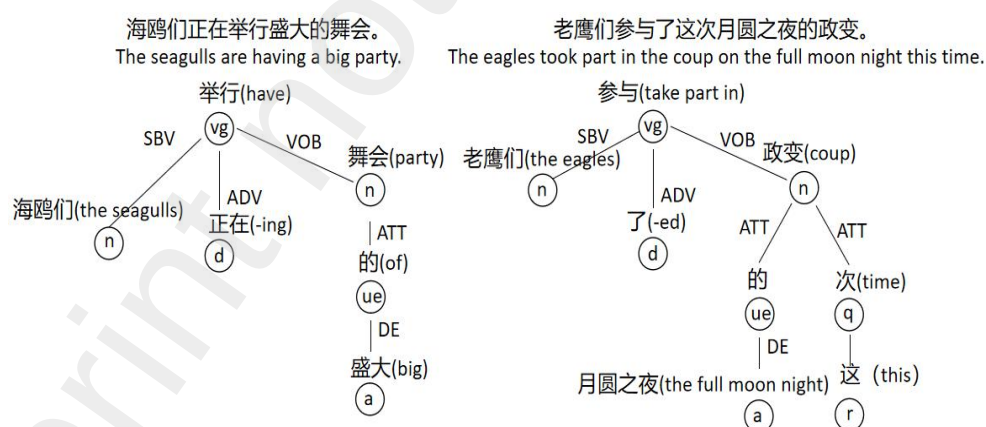
are expressed using sentences. If it is a proper noun representing a human-specific event, then the semantic features of the personification rhetoric single-sentences are not complex and primarily focus on the lexical level. However, if a sentence represents an event unique to humans, then its semantic features elevate to the sentence level⁴.

(21) *The seagulls are holding a grand ball.*

(22) *The eagles participated in the coup on the night of the full moon.*

(23) *The flying competition attracted the pigeons from the forest.*

To identify single-sentence with event personification, the first step is to recognize expressions specific to human-exclusive events (which may include human-exclusive event words or verb-object phrases describing such events). When a non-human noun acts as the subject, and the human-exclusive event word serves as the predicate or object, the phenomenon of event personification is deemed present, thereby classifying the sentence as an single-sentence with event personification. The dependency syntax analysis diagram for example sentences using event personification rhetoric is provided below:



⁴ Event personification and action personification intersect but are different; the main distinction lies in that events involve multiple individuals, while actions mostly involve a single individual.

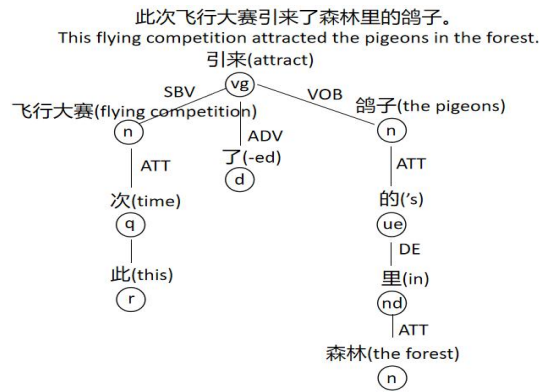


Figure 7 Dependency Syntax Analysis figure for Single-Sentence with event Personification

In event personification rhetorical sentences, some human-exclusive event nouns "N_{human-exclusive event}" appear in the subject position, while others appear in the object position. When "N_{human-exclusive event}" appears in the subject position, the object is required to be a non-human noun. Conversely, when "N_{human-exclusive event}" appears in the object position, the subject is required to be a non-human noun. Both these combinations are considered instances of event personification. Sentences containing this phenomenon are classified as event personification rhetorical sentences. From the dependency syntax analysis diagram of event personification rhetoric, The pattern framework of single-sentence with event personification can be summarized as follows:

Table 8 Semantic pattern Table of Single-Sentences with Event Personification

Category	Description	Framework	Example Sentence
1	"x SBV y" structure: x is a non-human noun, y is a human-exclusive event verb.		The eagles participated in the coup during the full moon night.
2	"x SBV y VOB z" structure: x is a non-human noun, y is a verb, z is a human-exclusive event noun.		The flight competition attracted pigeons from the forest.

Rule 10: if $\exists n_1 \in N_{\text{human-exclusive-event}}$, $\exists n_2 \in N_{\text{non-human}}$, $n_1 \in H_s$, $n_2 \in H_s$, and n_1 serves as the object of the sentence, and n_2 serves as the subject of the sentence, then $s \in S_{\text{event-personification}}$.

Rule 11: if $\exists n_1 \in N_{\text{human-exclusive-event}}$, $\exists n_2 \in N_{\text{non-human}}$, $n_1 \in H_s$, $n_2 \in H_s$, and n_1 serves as the subject of the sentence, and n_2 serves as the object of the sentence, then $s \in S_{\text{event-personification}}$.

3.3 Construction of Compound Sentence Interpretation Model

Compound-sentence personification identification is more complex than single-sentence analysis due to concealed semantic features. While compound sentences permit diverse semantic combinations, localized personification instances (e.g., in subordinate clauses) do not inherently classify the entire sentence as personification. When clauses are syntactically distant from the core structure, isolated anthropomorphic elements lack sufficient contextual influence to warrant global rhetorical categorization. Recognition requires evaluating proximity to main syntax and semantic integration rather than isolated markers, avoiding overgeneralization from peripheral figurative fragments. For example:

(24) *The seagull slowed its speed until the wind whispered in its ear, and until the ocean below became calm again.*

Through analysis, we can see that “the wind whispered in its ear” and “the ocean below became calm again” can both be identified as personification phenomena⁵. However, from a syntactic structural perspective, both of these parts are not core components of the sentence. Therefore, we cannot classify this sentence as a personification sentence based solely on these two parts⁶.

3.3.1 Coordinated Compound Sentence Interpretation Pattern

In Chinese, coordinated compound sentences consist of parallel simple-sentences linked by commas. Their core syntactic structures are equally parallel, with no hierarchical distinction. Consequently, personification identification in such sentences simplifies to detecting personification in individual clauses: if any constituent simple-sentence contains personification (e.g., anthropomorphic verbs/actions), the entire coordinated structure qualifies as personified. This approach avoids overcomplication by treating parallelism as semantic equivalence rather than requiring holistic rhetorical analysis. Example: “The wind whispered, leaves danced, and clouds wept” inherits personification status from each independent clause.

(25) *The sea breeze caresses the waves, the moonlight spills over the surface of the sea, the fish have all fallen into a deep sleep, and the seagulls are holding a grand*

⁵ The former being action personification and the latter being emotional personification.

⁶ Although this sentence contains instances of personification, it does not mean that the sentence itself is a personification sentence.

ball.

(26) *The eagles participated in the coup on this full moon night, the foxes have arrived, the coyotes have come, and the leopards are also here to support the eagles.*

In the examples of coordinated compound sentences with personification rhetoric above, when identifying complex rhetorical texts, it is sufficient to check whether any of the simple-sentences are personification sentences, such as (25) and (26).

3.3.2 Nested Compound Sentence Interpretation Model

Compared to coordinated compound sentences, nested compound sentences are more complex in their structure. The clauses forming a nested compound sentence not only have a coordinated relationship but may also include phenomena such as attributive clauses and adverbial clauses. When identifying personification in nested compound sentences, the following rules should be noted:

First, when there are nested subject-verb phrases, if two or more verbs appear along the dependency path from the sentiment word to the N_{animal} , it is not recognized as a personification sentence.

Second, when there are nested attributive phrases, if one or more nouns appear along the dependency path from the sentiment word to the animal noun, it is not recognized as a personification sentence.

Third, the grammatical distance between $N_{\text{non-human}}$ and the head word of the entire sentence should not exceed four steps.

(27) *The seagull slowed down until the wind whispered in its ear, until the ocean beneath it became calm again.*

This illustrates personification through a complement clause. The sentence is identified as personification primarily due to the combination "the seagull slowed down" rather than the expression of "in its ear" or "beneath it." Although those phrases also exhibit personification, their structural distance from the sentence's main verb "slowed" is too far. Check the dependency syntax analysis in the diagram below:

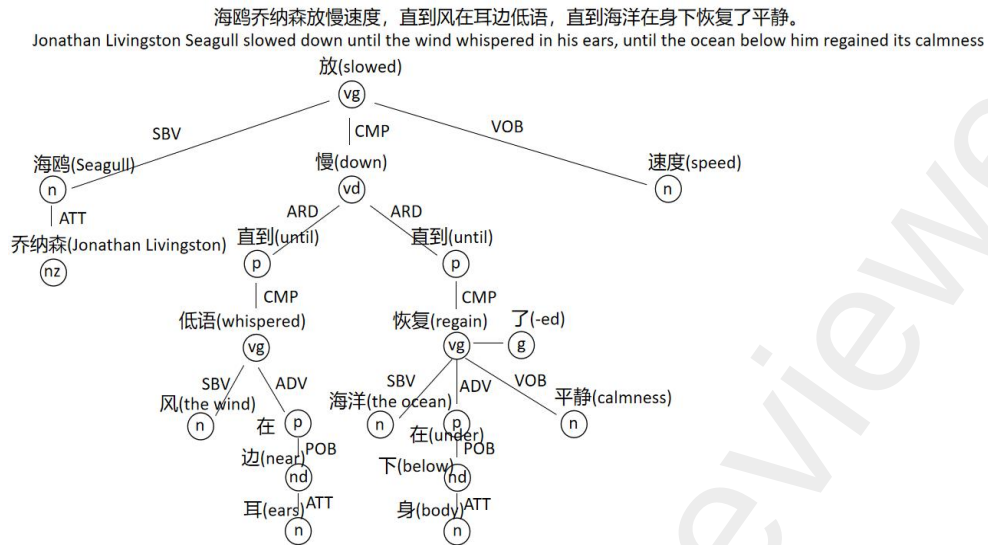


Figure 8 Dependency Syntax Analysis figure for Nested Compound Sentences

Regardless of which grammatical level the combination of " $N_{\text{animal}} + N_{\text{identity}}$ " appears at, it is identified as personification. In other words, as long as this combination is present, it is considered a phenomenon of personification. This characteristic often occurs at the lexical level, making it a relatively easy feature to identify.

4 Evaluation of Imagery Rhetoric Interpretation

Section 3 analyzes Chinese imagery rhetoric's semantic feature extraction and interpretative patterns, focusing on personification's generative mechanisms, recognition principles, and rhetorical efficacy through cognitive reprocessing of abstract concepts.

4.1 Evaluation of Personification Figure Recognition

The personification cognitive model, grounded in an autonomous-dependent framework, integrates conscious sensitivity principles to explain rhetorical generation and recognition. Generation begins with the writer's intent, driving cognitive-guided semantic analysis to identify correlations between the tenor (non-human entity) and vehicle (human traits). This involves inference, feedback, and syntactic integration per linguistic rules to construct personified expressions. Recognition reverses this process: syntactic decomposition and analysis of rhetorical rules reveal tenor-vehicle correlations, enabling reverse cognitive disassembly to reconstruct the writer's intent. Compared to Chapter 3's strategies, this model offers granular, structure-specific semantic analysis, prioritizing targeted identification of anthropomorphic features across sentence types. By systematically linking intent-driven generation to syntax-aware recognition, it bridges conceptual abstraction (tenor-vehicle mapping)

with formal linguistic constraints, enhancing precision in both creating and decoding figurative language.

4.2 Analysis of the Strengths of Personification Rhetoric

Semantic intuition told us that the effects of personification vary in strength. The strength of rhetorical effects is closely related to the syntactic hierarchy at which personification phenomena occur; the closer they are to the core syntactic structure, the more pronounced the personification effect is, and vice versa. For example, in the recognition rules for personification rhetoric in nested complex sentences, Rule 3 stipulates that the grammatical distance between a $N_{\text{non-human noun}}$ and the central term of the entire sentence must not exceed four steps. Under similar conditions, the first step is clearly more effective than the second step, the fourth step is weaker than the third step, and anything beyond four steps can no longer be identified as personification. From a syntactic perspective, Chinese personification rhetoric can be divided into two categories: personification based on the sentence as a semantic unit and personification based on syntactic components as semantic units. If we extract the core components of a sentence (subject + verb + object), the attribute of the subject can form personification with the subject, the adverbial and complement of the verb can form personification with the verb, and the attribute of the object can form personification with the object. In terms of sentence-level personification, the subject is a $N_{\text{non-human}}$, and the verb is a $V_{\text{human-related}}$, while there are no special requirements for personification at the component level.

From the perspective of human cognition, the strongest form of personification involves directly viewing an entity as a human. A slightly weaker form attributes human-like emotions to it, an even weaker form attributes human-like actions, and the weakest form is when only a part of it resembles a human. This aligns with the prototype theory proposed by cognitive semantics. The personification mentioned above pertains to different perspectives within a single entity. In contrast, relationships and events represent personification perspectives that extend beyond individual entities, making their degree of personification stronger than that of a single entity. In terms of scope, we believe that event personification is stronger than relational personification. After completing the recognition of personification in the first step, if we were to rank the degrees of personification, it would be as follows:

$$N_i \text{ Event} > N_i \text{ Relationship} > N_i \text{ Identity} > N_i \text{ Emotion} > N_i \text{ Action} > N_i \text{ Part}$$

5 Identifying Artistic Connotation in Rhetoric

This section selects the Ministry of Education's Chinese language textbooks for 11 grade levels in primary and secondary schools, as well as some classic prose passages, as experimental data. The data set for our research was created through manual annotation, resulting in a total of 2,329 pieces of annotated rhetorical text, including 1,329 pieces of personification rhetoric and 1,000 pieces of non-personification rhetoric. In addition to studying the binary classification problem of personification rhetoric, this chapter further completed a six-class classification annotation of personification rhetoric texts based on the classification system proposed in this chapter. The specific details of the annotations are as follows:

Table 9 Data Annotation Statistics for Personification Rhetoric Classification

lassification System	Identity/Title Personification	Characteristic Personification	Emotional Personification	Action Personification	Relationship Personification	Event Personification
Annotated Quantity (Pieces)	126	127	213	599	117	147
Annotation Code	1	2	3	4	5	6

5.1 Automated Recognition Contrast Experiment

Using the annotated experimental dataset mentioned above, we validate and evaluate the performance of the automated recognition model proposed in this paper. We design a binary classification contrast experiment for the automatic recognition of personification rhetoric, with comparative analysis of the experimental results shown in Table 9.

Experiment 1: Word-BiLstm-Attention [WBA] - This method involves feeding sentence feature vectors, composed of equal vocabulary, into a BiLstm network to learn contextual information of words, while also incorporating an attention mechanism to allocate different weights to words for joint learning of personification rhetoric recognition.

Experiment 2: Word-POS-BiLstm-Attention [WPBA] - In this method, the word vectors and part-of-speech vectors are fused to form a sentence vector matrix, which is then input into the BiLstm network to learn the contextual information of words. An attention mechanism is also added to allocate different weights to words for joint learning of personification rhetoric recognition.

Experiment 3: Word-Grammar-BiLstm-Attention [WGBA] - This method fuses word vector features and grammatical features to create a sentence vector matrix,

which is then fed into the BiLstm network to learn contextual information of words. An attention mechanism is also incorporated to assign different weights to words for joint learning of personification rhetoric recognition.

Experiment 4: Word-POS-Grammar-BiLstm-Attention [WPGBA] - In this approach, the word vector features, part-of-speech vectors, and grammatical features are combined to form a sentence vector matrix, which is then input into the BiLstm network for contextual learning of words, along with an attention mechanism to allocate different weights to words for joint learning of personification rhetoric recognition.

Experiment 5: Word-POS-Grammar-BiLstm [WPGB] - This method combines word vector features, part-of-speech vectors, and grammatical features to create a sentence vector matrix, which is then passed into the BiLstm network to learn the contextual information of words without including an attention mechanism for personification rhetoric recognition.

Experiment 6: Word-POS-Grammar-CNN-Attention [WPGCA] - In this method, the word vector features, part-of-speech vectors, and grammatical features are fused to form a sentence vector matrix, which is input into a CNN network to learn local features within the sentence. An attention mechanism is also included to allocate different weights to words for joint learning of personification rhetoric recognition.

This study investigates multi-class personification text recognition using annotated datasets, evaluates the proposed model's automatic classification capability, and designs validation experiments.

Table 10 Experiment for Automated Recognition of Personification Rhetoric Texts

No.	Multi-Class Strategy	Abbreviation	Description
1	[Word-BiLstm-Attention] ₇	WBA ₇	Uses only word vectors as feature input
2	[Word-POS-BiLstm-Attention] ₇	WPBA ₇	Combines word vectors and part-of-speech vectors
3	[Word-Grammar-BiLstm-Attention] ₇	WGBA ₇	Uses word vectors and grammatical features
4	[Word-POS-Grammar-BiLstm-Attention] ₇	WPGBA ₇	Multi-feature fusion of word vectors, part-of-speech, grammar, and semantics
5	[Word-POS-Grammar-BiLstm] ₇	WPGB ₇	No attention mechanism designed in the algorithm
6	[Word-POS-Grammar-CNN-Attention] ₇	WPGCA ₇	Uses CNN instead of BiLstm in the algorithm

Based on the previously mentioned multi-class experiment design for the automated recognition of personification rhetoric texts, each comparative experiment was conducted and the results were statistically compiled, as shown in Table 11.

Table 11 Analysis of Multi-Class Algorithms for Personification Rhetoric Recognition

NO.	Algorithm	Accuracy A (%)	Recall R (%)	F1 (%)
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1	WBA ₇	38.6	35.9	37.2
2	WPBA ₇	52.7	43.8	44.6
3	WGBA ₇	54.0	45.5	46.3
4	WPGBA₇	55.7	50.8	51.4
5	WPGB ₇	54.1	48.5	49.3
6	WPGCA ₇	47.6	46.2	45.8

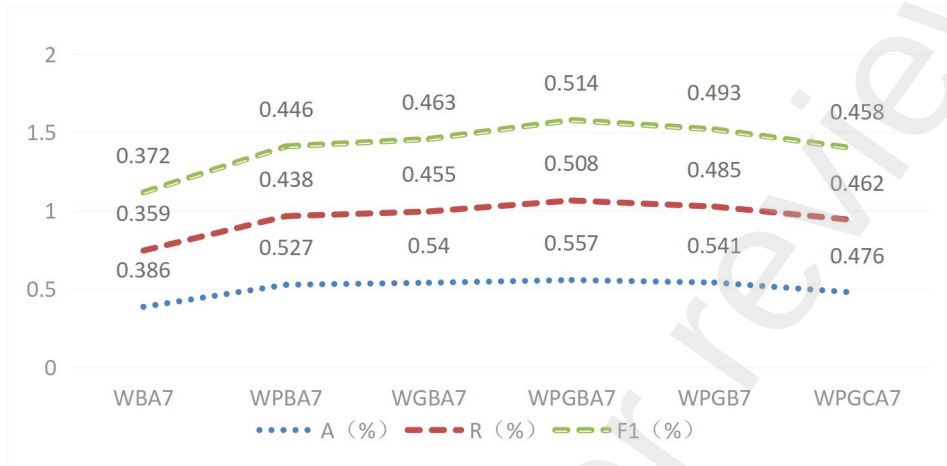


Figure 9 Results of Multi-class Experiment for Personification Rhetoric Recognition



Figure 10 Comparison of Results for Multi-class Experiment in Personification Rhetoric Recognition

This study proposes an anthropomorphic rhetoric recognition model by analyzing cognitive distinctions between anthropomorphic and metaphorical rhetoric. Corpus analysis reveals relationships between parts of speech and grammatical structures in anthropomorphic expressions. The method integrates Word2vec for semantic features and BiLSTM for contextual patterns, enhanced by Attention-based weight allocation. Multi-dimensional feature fusion enables automated detection, combining lexical meaning, syntactic context, and focused implicit characteristics through machine learning strategies.

Through the analysis of the experimental results in the above table, it can be known that the accuracy rate of the WBA₇ recognition method is the lowest (38.6%), and the accuracy rate of the WPGBA₇ recognition method is the highest (55.7%).

With the continuous increase of feature types, the accuracy rate, recall rate and F1 value of each fused algorithm show an upward trend, indicating that multi-feature fusion is also applicable in multi-classification recognition problems. Because natural language has a forward-backward dependency relationship in structure, only relying on convolutional neural networks to realize text multi-classification will ignore the context meaning of words.

6 Conclusion

This study advances Chinese rhetorical analysis by developing computational methods for recognizing metaphorical personification through cognitive-linguistic integration. We propose a dual-perspective framework combining single/compound-sentence classification with semantic-syntactic pattern analysis, establishing formal recognition criteria through dependency parsing and contextual feature extraction. A six-tier identification system was implemented, supported by an annotated dataset of contextual personification. Results demonstrate that merging cognitive models with semantic expressiveness significantly enhances automated recognition accuracy and AI's figurative language interpretation.

The research provides two key contributions: A structured taxonomy enabling domain-specific LLM training for rhetorical tasks, and Benchmark data facilitating educational applications in rhetoric analysis. These innovations bridge theoretical rhetoric with NLP implementations, driving the digital transformation of linguistic studies through context-aware processing solutions. The methodology demonstrates how cognitive-computational integration improves machines' capacity to decode complex figurative constructs while maintaining interpretative rigor, offering scalable approaches for advancing figurative language understanding in AI systems.

Conflict of Interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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